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Identification of gallbladder cancer by direct near-infrared measurement of raw bile combined with two-trace two-dimensional correlation analysis

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We demonstrated the utility of direct near-infrared (NIR) bile analysis for the identification of gallbladder (GB) cancer by employing two-trace two-dimensional (2T2D) correlation analysis to recognize dissimilar spectral features among diverse bile samples for potential improvement of discrimination accuracy. To represent more diverse clinical cases for reliable assessment, bile samples obtained from five normal, 44 gallstone, 25 GB polyp, six hepatocellular cancer (HCC), and eight GB cancer subjects were analyzed. Due to the altered metabolic pathways by carcinogenesis, the NIR spectral features of GB cancer samples, including intensity ratios of main peaks, were different from those of other sample groups. The differentiation of GB cancer in the principal component (PC) score domain was mediocre and subsequent discrimination accuracy based on linear discriminant analysis (LDA) was 88.5%. When 2T2D slice spectra were obtained using a reference spectrum constructed by the linear combination of the spectra of five pure representative bile metabolites and employed, the accuracy was improved to 95.6%. The sensitive recognition of dissimilar spectral features in GB cancer by 2T2D correlation analysis was responsible for the enhanced discrimination.

Introduction

Bile juice is a valuable disease-indicating specimen since its composition can vary depending on pathological conditions of related patients. We previously demonstrated the utility of a simple drop-and-dry near-infrared (NIR) measurement of raw bile to distinguish gallbladder (GB) cancer from gallstone and GB polyp.[1] Raw bile juice was dried directly on a non-NIR-absorbing polytetrafluoroethylene (PTFE) plate and transmission NIR spectra of the dried bile were acquired. The band shapes and relative band intensities of GB cancer samples were different from those of gallstone and GB polyp samples, thereby enabling the discrimination of GB cancer. This previously reported NIR bile analysis scheme has potential for clinical on-site analysis due to the simple and rapid analysis that requires no pretreatment of raw bile. Other than the analysis of bile, NIR spectroscopy has been extensively adopted for biomedical diagnosis[2,3] using relevant biological specimens including blood,[4,5] serum,[6–8] saliva,[9] and tissue,[10,11] and diverse chemometric and deep learning methods such as partial least squares (PLS),[5–8,12] linear discriminant analysis (LDA),[3,11,14] support vector machine (SVM),[13] and artificial neural network (ANN)[13] are combined for multivariate analysis of acquired NIR spectra.

This current study aims to strengthen the NIR scheme as a clinical screening tool by addressing two further important issues. First, the number of bile samples was increased to represent more diverse clinical cases for reliable assessment. To this end, 88 bile samples were prepared from five normal, 44 gallstone, 25 GB polyp, six hepatocellular cancer (HCC), and eight GB cancer subjects, increasing the number of bile samples by 2.6-fold than in our previous report (34 samples total; 20 gallstone, 10 GB polyp, and four GB cancer).[1] Here, bile samples of healthy (normal) subjects and HCC patients were included as important control and comparison groups, respectively, to examine whether the NIR scheme could be versatile for identification of HCC with a high occurrence rate.[15,16]

Second, the identification of GB cancer can be more challenging due to the wider variation of bile composition from the increased number of bile samples in this study. Effective recognition of minute spectral differences among the disease groups is crucial to secure discrimination accuracy. For this purpose, two-trace two-dimensional (2T2D) correlation analysis, which is able to sensitively highlight asynchronous spectral behaviors arising from dissimilar sample compositions, was employed.[17] In 2T2D correlation analysis, a reference spectrum is required for comparison to a sample spectrum. In this study, a reference spectrum was uniquely constructed by linear

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combination from the pure spectra of five main representative metabolites in bile: lecithin (phosphatidylcholine), cholesterol, glycodeoxycholate (GDCA, glycine-conjugated bile salt), taurodeoxycholate (TDCA, taurine-conjugated bile salt), and albumin.

The spectrum of each bile sample and corresponding calculated synthetic spectrum was used as sample $s(\nu)$ spectrum and reference spectrum $r(\nu)$, respectively, for 2T2D correlation analysis, and an asynchronous spectrum, $\Psi(\nu_1, \nu_2)$, which is a quantitative comparison of the spectral intensity variations observed at two different spectral variables, was obtained using eqn (1).

$$\Psi(\nu_1, \nu_2) = \frac{1}{2}[s(\nu_1) \cdot r(\nu_2) - r(\nu_1) \cdot s(\nu_2)] \quad (1)$$

Then, principal component analysis (PCA) was performed using the obtained 2T2D slice spectra of samples and distributions of subsequent principal component (PC) scores were analyzed to evaluate the ability of disease discrimination. Finally, using the PC scores as inputs, LDA was performed to assess accuracy for the discrimination of GB cancer *vs.* other diseases.

Experimental section

Collection of bile samples and acquisition of NIR spectra

The purpose and content of the research was thoroughly explained to each subject, and this study was performed under the guidelines of the Institutional Review Board of Hanyang University (HYI-18-227-3). With the consent of each subject, all bile samples were collected after cholecystectomy at Hanyang University Hospital (Seoul, Korea). The bile samples were immediately frozen after acquisition and stored below −20 °C to prevent further denaturation. A total of 88 bile samples was provided from five normal, 44 gallstone, 25 GB polyp, six HCC, and eight GB cancer subjects. Each bile sample varied in color from yellow-brown, to green, to dark green.

NIR spectra (resolution: 4 cm^{-1}) of all the raw bile samples were acquired (128 scans) by the same procedure as previously described.[1] Briefly, each sample was dropped on a PTFE plate, dried, and the corresponding transmission NIR spectrum was collected by fully covering the dried bile (approximately 5 mm in diameter) using a FT-NIR spectrometer (ABB, Canada). The NIR spectra of pure cholesterol, GDCA, TDCA, albumin (Sigma-Aldrich), and lecithin (Daejung Chemicals & Metals, Korea) were collected with the same spectrometer. All the spectral analyses including baseline correction, PCA, 2T2D correlation analysis, and LDA were executed using MATLAB R2016a (The MathWorks Inc., USA).

Results and discussion

NIR spectral features of bile samples and examination of spectral differences

Fig. 1(a) shows the average normalized NIR spectra (4500–4000 cm^{-1} range) of dried bile samples from the normal (light blue), gallstone (red), GB polyp (blue), HCC (black), and GB cancer (green) subjects. Each average spectrum is properly offset for easy visual comparison of the peak shapes. Before averaging the spectra in each group, the baseline of each sample spectrum was corrected to zero at 4500 cm^{-1}, and the baseline-corrected spectrum was divided by the corresponding peak area under the 4500–4000 cm^{-1} range for normalization. The superimposed shades with each average spectrum correspond to the standard deviations of absorbance at each wavenumber, indicating the degree of sample-to-sample spectral variation in each group. The average spectra without the shades are also displayed at the bottom.

In all five average spectra, the major bands appear around 4334, 4258, 4170, and 4046 cm^{-1}. These bands originated from the major bile components such as GDCA, TDCA, lecithin, cholesterol, and protein. The NIR spectra of these components (albumin to represent protein) are also shown in Fig. 1(e) for comparison. The peak shapes for normal, gallstone, GB polyp, and HCC samples are quite similar to each other, while dissimilar to those of GB cancer. In the case of GB cancer, the intensity ratio between the 4334 and 4258 cm^{-1} bands ($I_{4334/4258}$) is different from the other disease groups. The combination of CH_3 asymmetric stretching and CH_2 bending is responsible for the 4334 cm^{-1} peak and the 4258 cm^{-1} peak originates from the combination of CH_3 symmetric stretching and CH_2 bending.[18,19] Also, absorbance of the 4046 cm^{-1} peak corresponds to C–N–C amide protein vibration is higher in the GB cancer spectrum. In our former report,[20] the concentration determination of bile salts and lipidomic profiling for nine lipid classes in the bile samples were accomplished using LC-MS, and the relative abundance of bile salts as well as phosphatidylcholines were different in GB cancer samples. The dissimilar bile compositions were probably from the altered metabolic pathways by GB cancer and thereby reflected in the collected NIR spectra as shown.

For quantitative evaluation, $I_{4334/4258}$ values in the sample spectra of each group were calculated, and the corresponding average and standard deviation (vertical bar) were compared as shown in Fig. 1(b). The difference of $I_{4334/4258}$ values between the two comparison groups was statistically evaluated using the Mann–Whitney U test and all 10 combinatorial cases were tested. Groups showing significant differences are designated by *, **, or ***, corresponding to p-values smaller than 0.05 (95.0% confidence interval (CI)), 0.01 (99.0% CI), or 0.001 (99.9% CI), respectively. The $I_{4334/4258}$ values of GB cancer samples are higher than those of normal, gallstone, GB polyp, and HCC samples with the CIs of 95.0, 99.9, 99.9, and 99.0%, respectively. On the other hand, the $I_{4334/4258}$ values among the remaining four groups are not different. When the intensity ratios between the 4258 and 4046 cm^{-1} peaks ($I_{4258/4046}$) in each group were compared (Fig. 1(c)), similar results were observed. For example, the lower $I_{4258/4046}$ values of GB cancer relative to those of the remaining samples are significant, while the differences of $I_{4258/4046}$ values among the remaining four groups remain insignificant.

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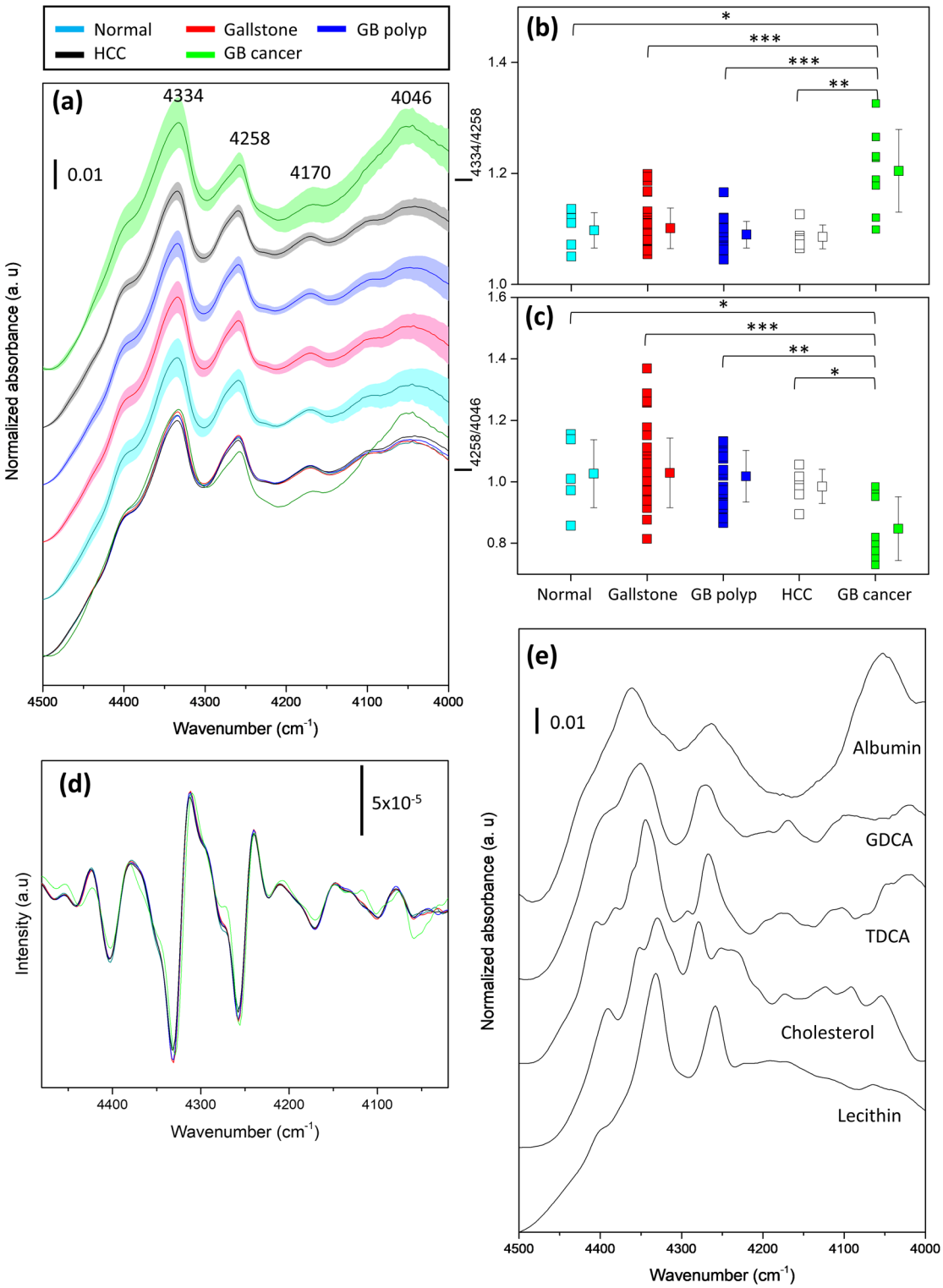

Fig. 1 Average NIR spectra (4500–4000 cm^{-1} range) of normal, gallstone, GB polyp, HCC, and GB cancer samples (a), and NIR spectra of GDCA, TDCA, lecithin, cholesterol, and albumin (e). The superimposed shaded areas with each average spectrum correspond to the standard deviation of absorbance at each wavenumber. $I_{4334/4258}$ (b) and $I_{4258/4046}$ (c) values with corresponding averages and standard deviations, and average second derivative spectra of the five groups (d) are also shown.

To examine the discriminability of GB cancer samples against the other samples, PCA was performed using all the sample spectra and subsequent PC scores were analyzed. Fig. 2(a) shows the distribution of first and second PC scores of the samples. The percentages of spectral variation explained by the respective PCs are denoted in parentheses, and these

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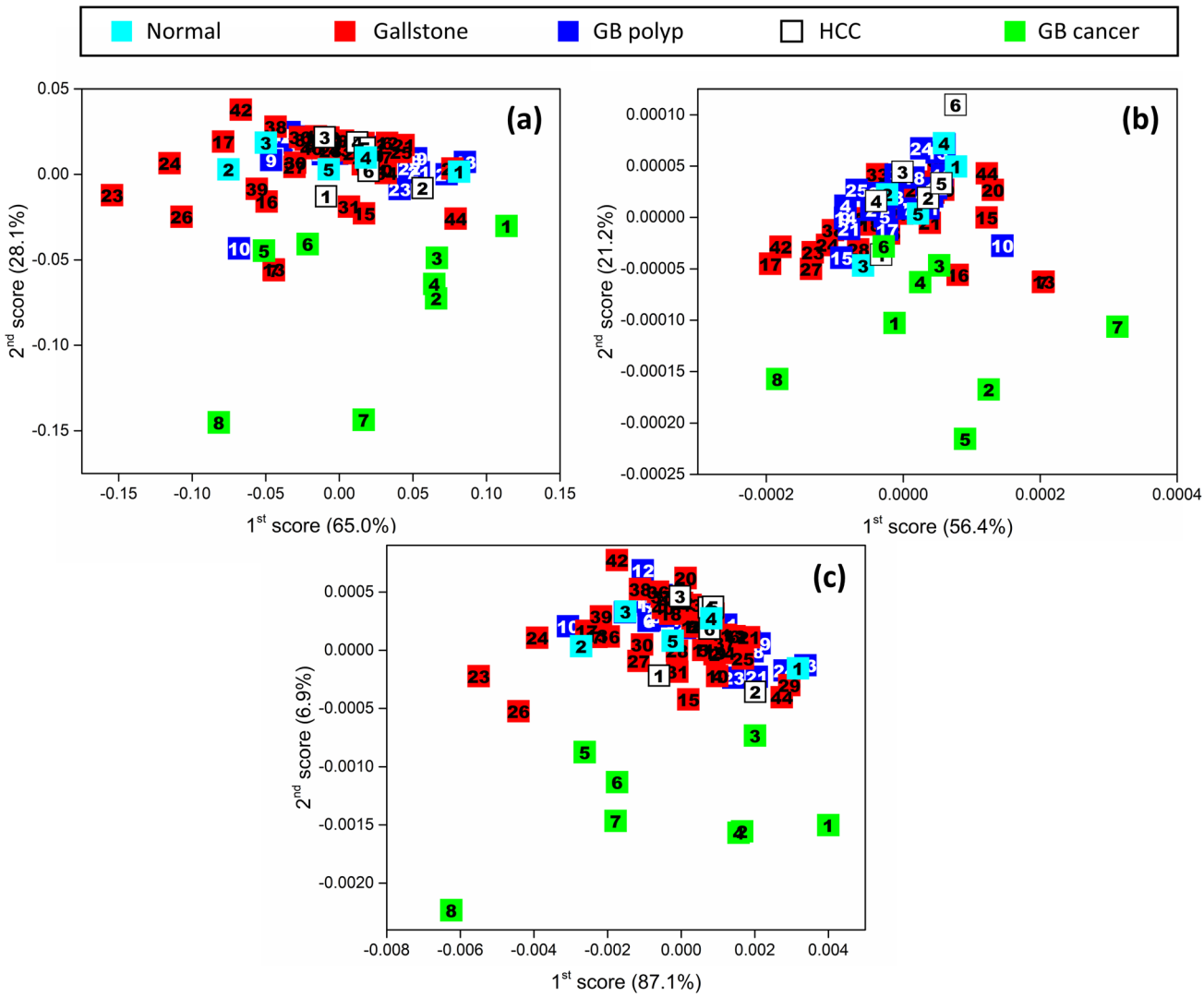

Fig. 2 Distributions of the first *vs.* second PC scores obtained from raw spectra (a), second derivative spectra (b), and 2T2D slice spectra at 4054 cm^{-1} (c) of all samples.

two PCs described 93.1% of the total spectral variation. The GB cancer group is crudely separable from the remaining groups, but not clearly differentiated. GB cancer samples #5 and #6 are within a cluster of the remained samples, and GB cancer sample #3 is close to the cluster. The distinction of HCC samples is impossible, indicating that the occurrence of HCC does not significantly alter the composition of bile.

To estimate accuracy for the discrimination of GB cancer, LDA was executed using the first three PC scores (describing greater than 96.0% of the total spectral variation) of the samples. Then, the normal, gallstone, GB polyp, and HCC samples were merged into one group for two-group (non-GB cancer and GB cancer) discrimination. The merge of the four groups resulted in unbalanced sample numbers between the non-GB cancer and GB cancer groups, 80 and eight samples, respectively, and subsequent discrimination accuracy could be misleading. For example, if all the 88 samples are classified as non-GB cancer, the accuracy is still good (90.9%) although all eight GB cancer samples are incorrectly classified.

To avoid this problem, five and three samples were randomly chosen from each group and assigned to a training and validation set, respectively, such that 10 samples were included in the training set and six samples were in the validation set. With these samples, LDA accuracy was evaluated. Then, out of the remaining 72 non-GB cancer samples, five training and three validation samples were randomly selected. Also, out of the eight GB cancer samples, five training and three validation samples were randomly chosen again. Using the newly selected samples, discrimination accuracy was obtained. The sample selection continued 10 times until all 80 non-GB cancer samples were chosen without duplication. The entire procedure was repeated 1000 times for cross-validation. Table 1 shows the LDA results including accuracy, sensitivity, and specificity. The accuracy is mediocre at 88.5%, as expected

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Table 1 Accuracy, sensitivity, and specificity of GB cancer discrimination when PC scores of raw, second derivative, and 2T2D slice spectra of raw spectra are used for LDA

Inputs for LDA	Accuracy (%)	Sensitivity (%)	Specificity (%)
PC scores of raw spectra	88.5 ± 4.0	96.2	80.9
PC scores of second derivatives	84.1 ± 4.1	91.1	77.1
PC scores of 2T2D slice spectra	95.6 ± 2.7	98.8	92.4

since the scores of the two groups are somewhat overlapped (Fig. 2(a)). Meanwhile, when the first six PC scores (describing greater than 99.0% of the total spectral variation) were used, the accuracy was decreased to 83.2%.

To potentially highlight the minute spectral differences among the samples, the corresponding second derivative spectra were also employed for the discriminant analysis. Fig. 1(d) shows the average second derivative spectra corresponding to each group. Compared to the raw spectra, the more diverse narrow peaks are observed in the second derivative spectra of five groups. Again, the second derivative peak shapes of normal, gallstone, GB polyp, and HCC sample groups are analogous to each other, while different from those of GB cancer. In the second derivative spectrum of GB cancer, the peaks different from those of the remained groups are noticed across the spectral range.

Fig. 2(b) presents the distribution of first and second PC scores obtained from the second derivative spectra of all the samples. These two PCs described 77.6% of the total spectral variation. The distinction of GB cancer samples from the remained samples is similar or not improved compared to that in the PC score domain based on the raw spectra. For example, #3, #4, and #6 GB cancer samples are within the cluster of the remained samples. When the first three PC scores (describing greater than 84.7% of the total spectral variation) were used for LDA, the accuracy was maximized to 84.1% (refer to Table 1), while inferior to 88.5% using the PC scores of raw spectra. When the first six PC scores (describing greater than 95.0% of the total spectral variation) were used, the accuracy was substantially degraded to 68.4%. Since the raw spectra of GB cancer contained the sufficiently dissimilar spectral features for the discrimination, the utility of second derivatization

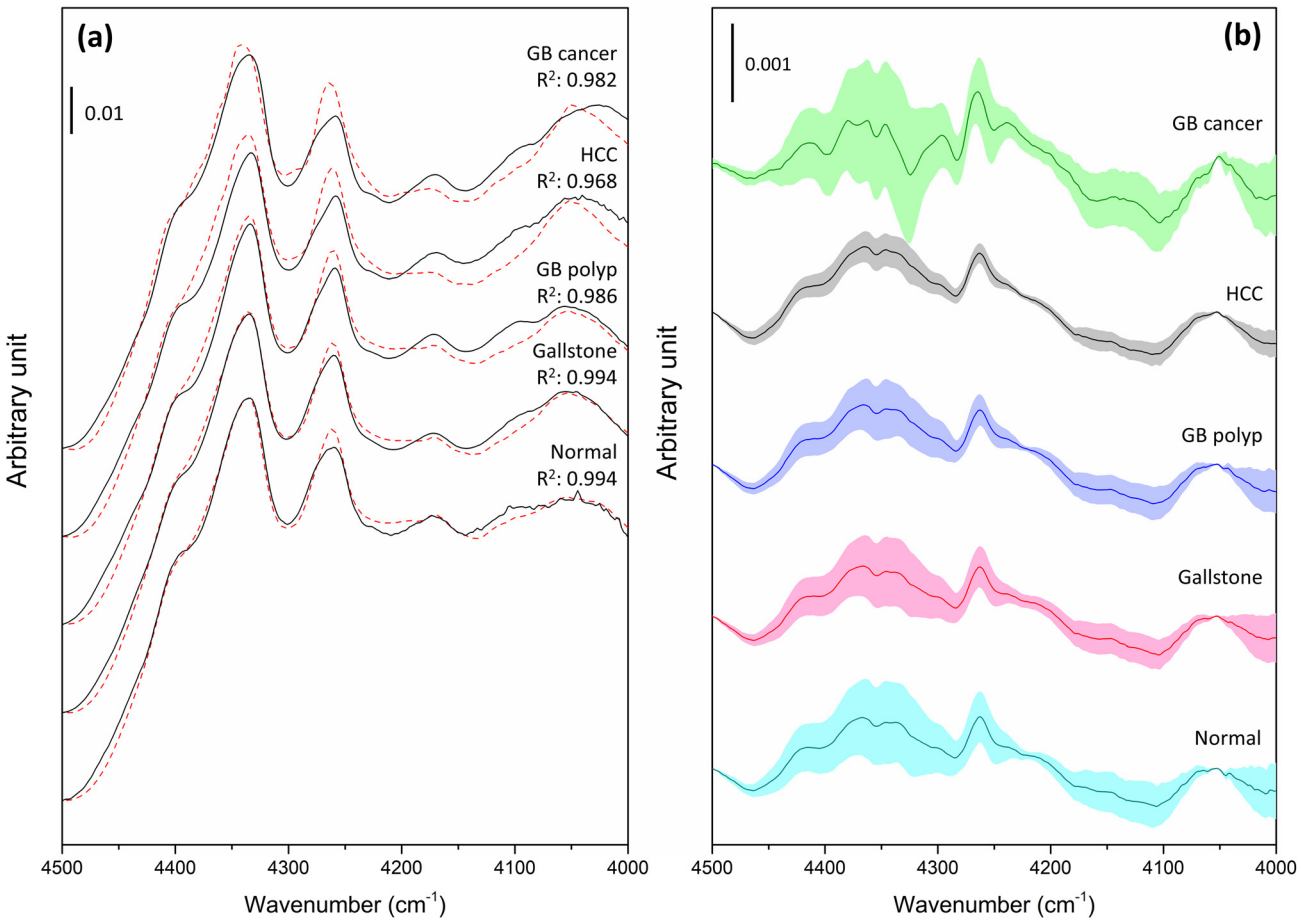

Fig. 3 NIR spectra (solid line) of one sample chosen from each of the five groups and corresponding reference spectra (red dashed line) constructed by regression of the five pure bile metabolite spectra (a), and the average 2T2D slice spectra at 4054 cm^{-1} of the normal, gallstone, GB polyp, HCC, and GB cancer samples (b).

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was not readily realized in this case. Therefore, the raw spectra of samples were solely employed for subsequent 2T2D correlation analysis.

2T2D correlation analysis and use of 2T2D slice spectra for GB cancer discrimination

As a means of improving the discrimination, 2T2D correlation analysis[17] was adopted to recognize minutely dissimilar NIR spectral features among the samples. To execute 2T2D correlation analysis, a reference spectrum was necessary and was produced by linear combination (regression) of the five pure metabolite spectra (Fig. 1(e)) for each sample spectrum. The assumption behind the linear combination was that there were no interactions among the components. Fig. 3(a) shows the spectra of one chosen sample from each of the five groups (solid line) and corresponding reference spectra (dashed line) constructed by the regression. The resulting R^2 values for each case are designated. As seen, the reference spectra closely resemble the corresponding sample spectra, although they could not fully represent the spectra of real bile samples due to the existence of inter-component interactions.

Fig. 4(a), (b), (c), (d), and (e) shows the average asynchronous 2T2D correlation maps of the normal, gallstone, GB polyp, HCC, and GB cancer samples, respectively. The asynchronous correlation features of normal, gallstone, GB polyp, and HCC are quite similar with each other, while apparently dissimilar to that of GB cancer over the whole range. The presence of diverse narrow asynchronous peaks in the 4450–4250 cm^{-1} range is notable for GB cancer. This confirms that 2T2D correlation analysis recognizes the characteristic spectral features of GB cancer, which are dissimilar to those of the remaining four groups. Fig. 3(b) displays the average 2T2D slice spectra at 4054 cm^{-1} of normal, gallstone, GB polyp, HCC, and GB cancer samples. The shades superimposed with each average spectrum correspond to the standard deviations. The overall shapes of 2T2D slice spectra among the four non-GB cancer groups are similar, while greatly different from that of the GB cancer group, especially in the 4450–4250 cm^{-1} range. Also, the magnitude of shade along the 2T2D slice spectrum of the GB cancer spectrum is large, indicating substantial variation of bile composition.

PCA was performed again using the 2T2D slice spectra at 4054 cm^{-1} of all samples, and Fig. 2(c) presents the distribution of the first *vs.* second PC scores. These two PCs described 94.0% of the total spectral variation, and separation of the GB cancer group is apparently improved compared to using the raw spectra. GB cancer samples #5 and #6 positioned within the cluster of the remaining samples in Fig. 2(a) are now located apart from the non-GB cancer cluster. Only GB cancer sample

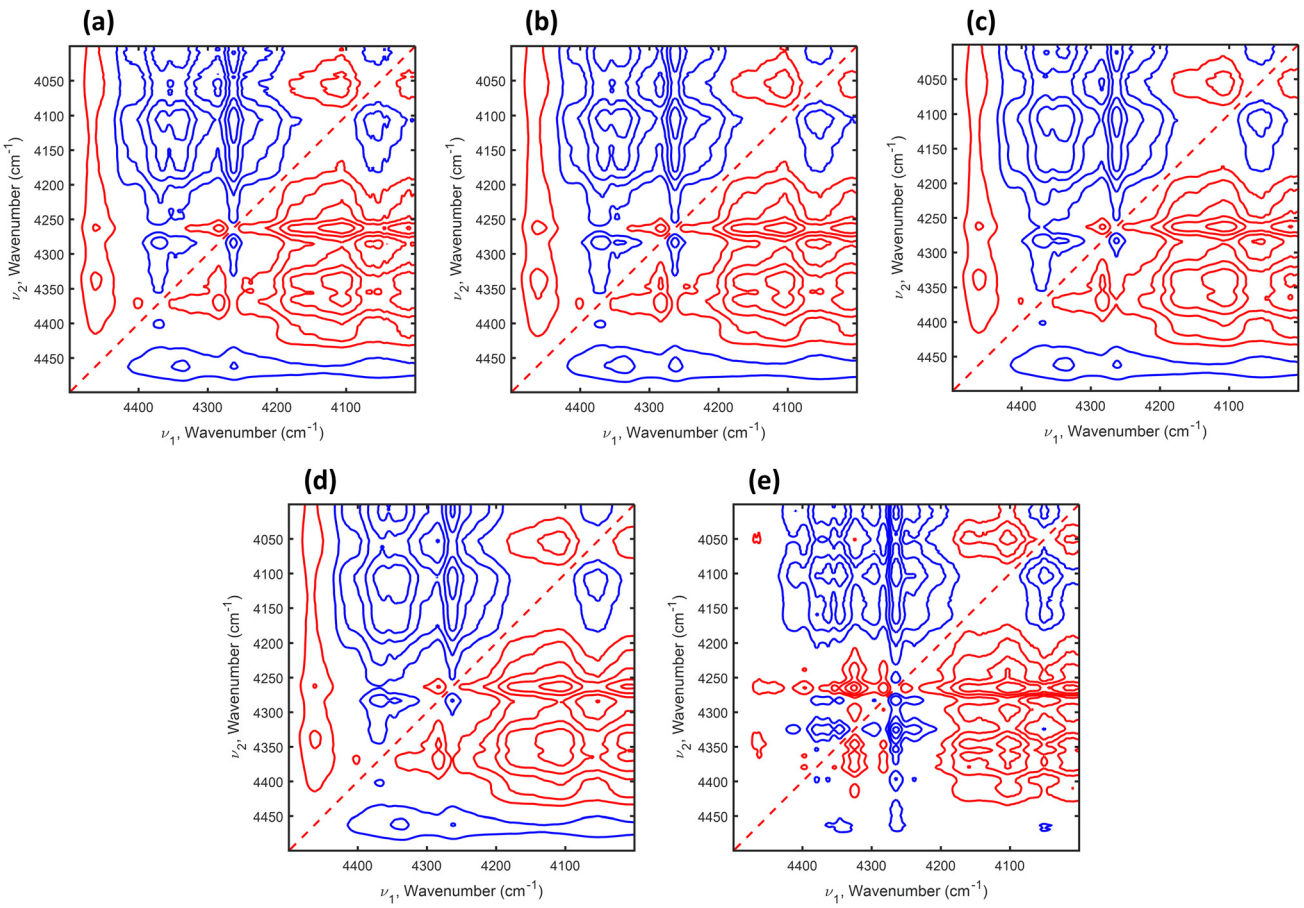

Fig. 4 Average asynchronous 2T2D correlation maps of the normal (a), gallstone (b), GB polyp (c), HCC (d), and GB cancer (e) samples.

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#3 is near the non-GB cancer cluster. Therefore, accuracy using the first three PC scores (describing greater than 96.8% of the total spectral variation) of the 2T2D slice spectra was considerably elevated to 95.6% and the standard deviation decreased to 2.7%, indicating a more robust discrimination performance. In particular, the improvement of specificity (identification rate of GB cancer) was notable (Table 1). When the first five PC scores (describing greater than 99.0% of the total spectral variation) were used, accuracy was again decreased to 91.0%.

Conclusion

The NIR spectral features, in particular the $I_{4334/4258}$ and $I_{4258/4046}$ values, of GB cancer samples are statistically different from those of normal, gallstone, GB polyp, and HCC samples. The varied bile salt and lipid composition by the occurrence of GB cancer was clearly reflected in the NIR spectra of bile samples. By using 2T2D slice spectra, the ability of GB cancer discrimination was improved with an accuracy of 95.6%, which is higher than 88.5% when employing the raw spectra alone. 2T2D correlation analysis is effective to recognize minutely dissimilar spectral features of GB cancer samples.

Although the number of total examined samples was increased in this study, a sample set including balanced sample distribution among the disease groups is still required for more reliable evaluation and clinical application. In particular, larger numbers of normal and GB cancer samples are necessary. While additional sample collection is underway, acquisition is slow since normal bile samples are only obtainable from healthy organ donors for liver transplantation and GB cancer is not a disease with a high incidence rate. When a balanced sample set with many samples is available, diverse deep learning methods will be evaluated to secure discrimination accuracy.

Author contributions

Eunjin Jang: investigation, methodology, visualization, formal analysis, writing – review & editing. Woosuk Sohng: formal analysis, software, validation. Donho Choi: conceptualization, resources. Hoeil Chung: conceptualization, project administration, funding acquisition, writing – original draft, writing – review & editing, supervision.

Conflicts of interest

There are no conflicts to declare.

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